Predicted Signal Magnitudes for Tabletop Tests of Emergent Vacuum Response Theory

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This work provides phenomenological estimates and simple numerical simulations of expected signal magnitudes for tabletop experimental tests of Emergent Vacuum Response Theory (EVRT). Building on previously proposed low-cost resonator-based measurement protocols, we model small deviations from standard electromagnetic cavity decay in terms of fractional changes in ringdown time, phase response, and effective linewidth. The objective is not to predict specific microscopic mechanisms, but to establish order-of-magnitude expectations and signal-to-noise considerations for experimentally accessible systems. These estimates provide practical guidance for interpreting null results and identifying candidate deviations in moderate-Q resonator experiments.

I. INTRODUCTION

Emergent Vacuum Response Theory (EVRT) proposes that coherent nonequilibrium electromagnetic systems may exhibit small, resonance-dependent deviations from conventional linear electromagnetic response. Previous work has outlined both the theoretical framework and a minimal-cost experimental protocol based on tabletop resonator systems.

A key remaining question is the expected magnitude and structure of measurable signals under realistic laboratory conditions. The purpose of this work is to provide simple phenomenological models and numerical estimates that connect experimental observables to plausible deviation scales, without assuming a specific microscopic origin.

II. PHENOMENOLOGICAL MODEL

We model deviations from standard exponential ringdown behavior using a perturbative modification to the decay constant:

$$A(t) = A_0 e^{-t/\tau_{\text{eff}}}, \tag{1}$$

where

$$\tau_{\text{eff}} = \tau_0(1+\epsilon), \tag{2}$$

and $\epsilon \ll 1$ represents a small fractional deviation from the Maxwellian expectation.

The corresponding observable is:

$$\frac{\Delta\tau}{\tau_0} = \epsilon. \tag{3}$$

This framework allows direct comparison with experimental measurements of ringdown time.

III. SIMULATION FRAMEWORK

To estimate detectability, we simulate ringdown curves with small deviations superimposed on realistic noise.

The measured signal is modeled as:

$$A_{\text{meas}}(t) = A_0 e^{-t/\tau_0(1+\epsilon)} + \eta(t), \tag{4}$$

where $\eta(t)$ represents measurement noise, modeled as Gaussian white noise with variance σ^2.

Representative simulation outputs are shown in Figs. 1–4.

Simulations are performed across a range of:

- fractional deviations $\epsilon \sim 10^{-6}$ to 10^{-2},
- quality factors $Q \sim 10^3$–10^5,
- sampling rates and noise levels representative of typical oscilloscopes.

IV. PREDICTED SIGNAL MAGNITUDES

For a resonator with frequency $f \sim 100\,\text{MHz}$ and $Q \sim 10^4$, the baseline ringdown time is:

$$\tau_0 \sim \frac{Q}{2\pi f} \approx 16\,\mu\text{s}. \tag{5}$$

Representative deviations yield:

- $\epsilon \sim 10^{-3} \Rightarrow \Delta\tau \sim 16\,\mathrm{ns}$,
- $\epsilon \sim 10^{-4} \Rightarrow \Delta\tau \sim 1.6\,\mathrm{ns}$,
- $\epsilon \sim 10^{-5} \Rightarrow \Delta\tau \sim 0.16\,\mathrm{ns}$.

These values are at or near the resolution limits of standard high-bandwidth measurement systems.

V. SIGNAL-TO-NOISE CONSIDERATIONS

Detectability depends on the ratio:

$$\mathrm{SNR} \sim \frac{\Delta\tau}{\sigma_\tau}, \tag{6}$$

where σ_τ represents uncertainty in extracted ringdown time.

Monte Carlo simulations indicate that:

- $\epsilon \gtrsim 10^{-3}$ is typically resolvable with modest averaging,
- $\epsilon \sim 10^{-4}$ requires careful noise control and repeated trials,
- $\epsilon \lesssim 10^{-5}$ is likely below the noise floor for basic setups.

VI. ADDITIONAL OBSERVABLES

In addition to ringdown time, EVRT-like deviations may manifest as:

- phase lag between drive and response,
- resonance linewidth broadening or narrowing,
- amplitude-dependent shifts in effective Q.

These effects may provide complementary detection pathways.

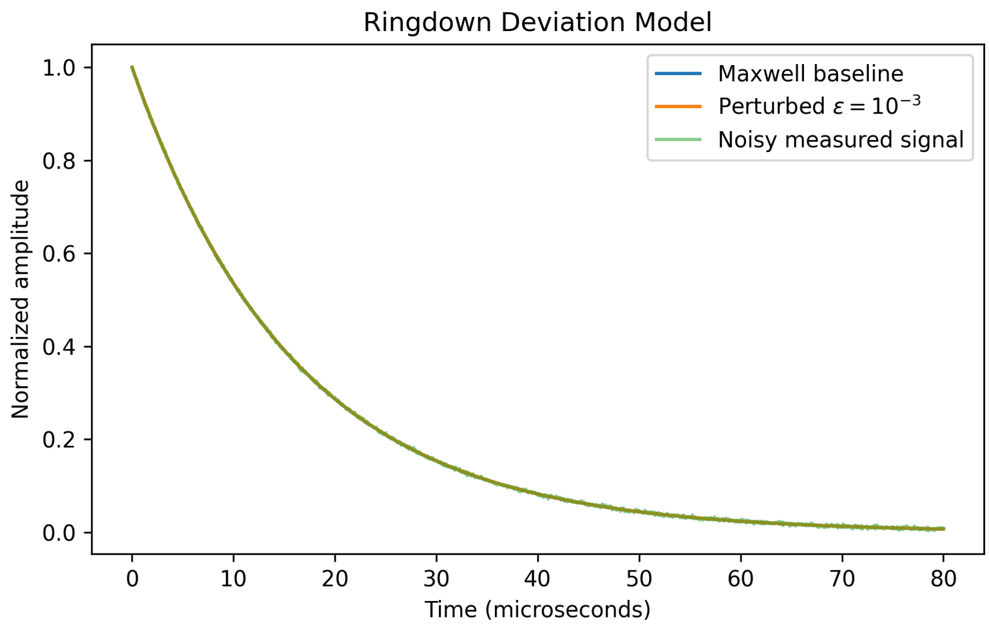

FIG. 1. Illustrative ringdown comparison between a Maxwellian baseline, a small perturbed decay with fractional deviation $\epsilon = 10^{-3}$, and a noisy measured trace. The purpose of the plot is to show the qualitative scale of deviations that a tabletop test would attempt to resolve.

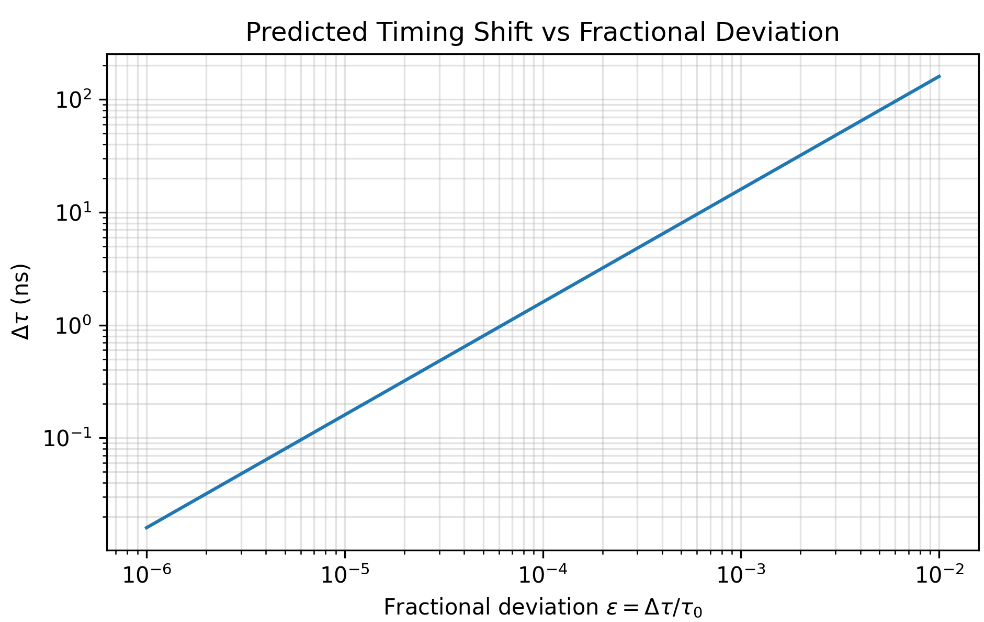

FIG. 2. Predicted timing shift $\Delta\tau$ as a function of fractional deviation $\epsilon = \Delta\tau/\tau_0$ for a representative baseline ringdown time $\tau_0 = 16$ μs.

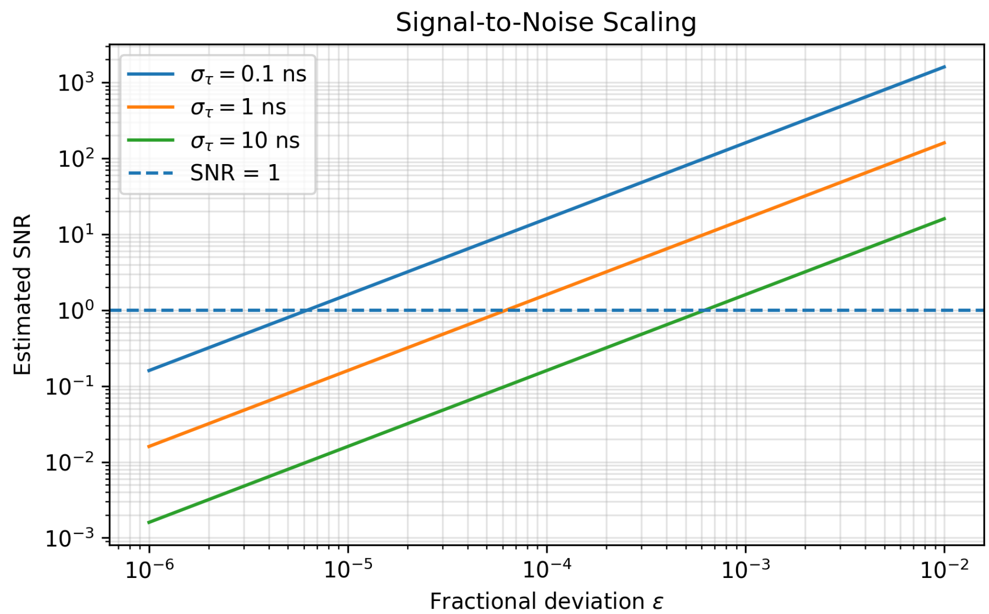

FIG. 3. Estimated signal-to-noise ratio as a function of fractional deviation for representative ringdown-time uncertainties. The transition near SNR ~ 1 indicates the approximate lower boundary for reliable detection without additional averaging or improved timing stability.

VII. SIMULATION FIGURES AND DETECTABILITY MAPS

The following figures provide illustrative numerical outputs for the simplified signal model. These plots are not presented as evidence for EVRT, but as practical guides for identifying which measurement regimes could plausibly distinguish small deviations from ordinary resonator decay.

These results directly inform the experimental protocol described in prior work by identifying the parameter regimes in which measurable deviations would be most likely to emerge.

VIII. INTERPRETATION

These estimates do not assume any specific physical mechanism, but instead define a practical parameter space for experimental testing.

A null result at sensitivity $\epsilon \sim 10^{-3}$–10^{-5} places quantitative constraints on any EVRT-like coupling under accessible conditions. Conversely, any reproducible deviation within this range would warrant further investigation using higher-precision instrumentation.

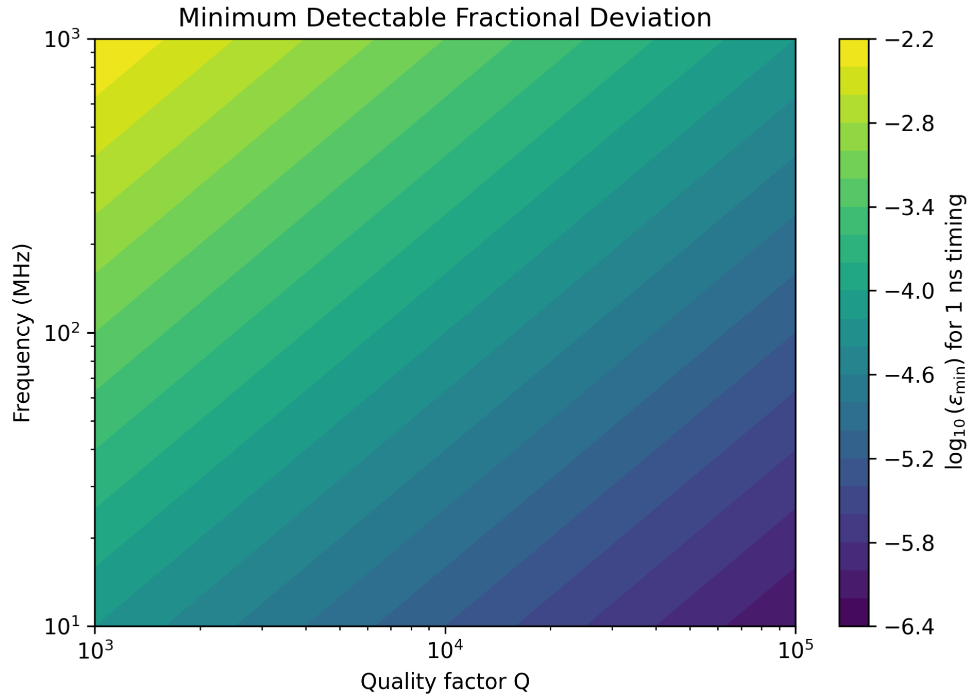

FIG. 4. Minimum detectable fractional deviation ϵ_{min} for a representative timing uncertainty of 1 ns as a function of resonator quality factor and frequency. Higher quality factors and lower frequencies generally increase ringdown duration and improve fractional timing sensitivity.

IX. CONCLUSION

This work provides order-of-magnitude estimates for expected signal magnitudes in tabletop tests of EVRT. By connecting phenomenological deviations to experimentally measurable quantities, these results complement existing experimental protocols and help define realistic detection thresholds.

Together with prior theoretical and experimental work, this establishes a complete and testable framework for probing potential nonequilibrium electromagnetic response in resonant systems.

REFERENCES

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